

# Decomposition of Concentration-Index Trees

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## 1 Results

### 1.1 Exploratory Data Analysis

In the DRC Data Set ([The DRC Program Program, 2007](#)), 10.32% of children died before reaching age five (Figure 1). Deaths were more commonly observed among children from disadvantaged or rural backgrounds. The children who died differed descriptively from those who survived. Low-wealth households accounted for a larger share of deaths than of survivors (51.3% versus 42.8%). The same pattern was present for rural residence (69.3% versus 60.7%), no skilled attendance at birth (65.6% versus 56.8%), and mothers with no education (67.5% versus 55.2%) (Table 2). Mortality also differed across provinces, ranging from 6.2% in Kinshasa to 14.2% in Katanga (Figure 12). Next we use the direct and indirect methods for decomposition.

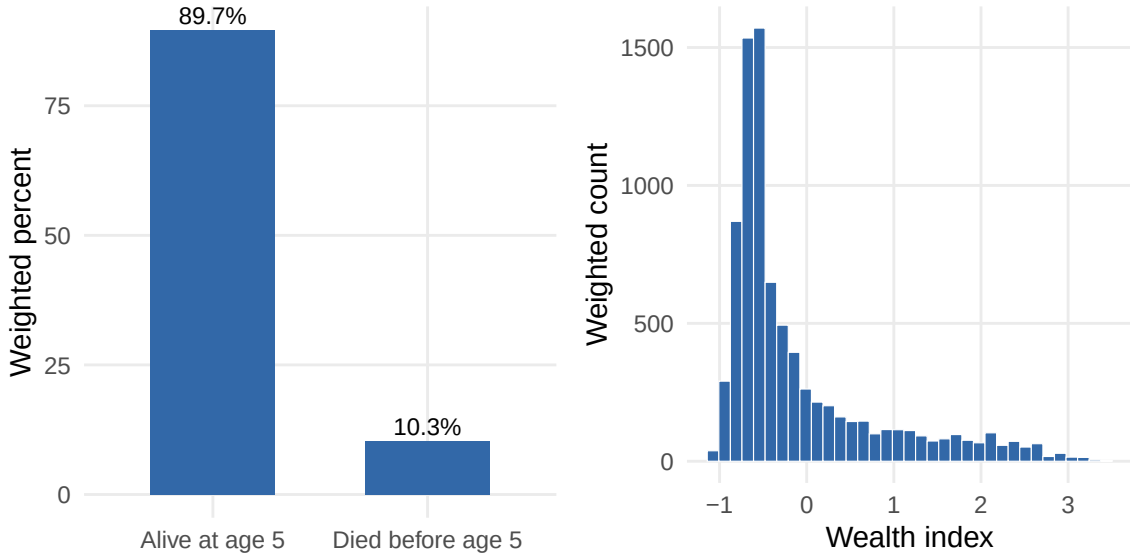


Figure 1: Under-five mortality distribution and DHS wealth index in the DRC analysis data.

### 1.2 Decomposition of determinants of under-five mortality

#### 1.2.1 Indirect methods comparison (regression and predictive-forest SHAP)

From the indirect decomposition, we see that the largest regression contribution came from mother’s education: 23.1% under CI and 28.7% under CIg/CIc (Table 1). Father’s occupation in agriculture was second, contributing 17.9% under CI and 19.9% under CIg/CIc. In the predictive-forest SHAP decomposition, the contributions were less concentrated in one or two variables.

Residence and household wealth each contributed about 12% under CI, CIg and CIc, followed by mother’s occupation, father’s occupation, mother’s education and skilled birth attendance. The fitted regression model is reported in Table 3. Details of the fitted predictive forest are provided in Table 4 and Table 5, and its SHAP contributions are shown in Figure 13.

Table 1: Regression and predictive-forest SHAP percentage contributions by criterion.

| Variable                                                     | Regression |         |         | Predictive-forest SHAP |         |         |        |
|--------------------------------------------------------------|------------|---------|---------|------------------------|---------|---------|--------|
|                                                              | CI (%)     | CIg (%) | CIc (%) | CI (%)                 | CIg (%) | CIc (%) | L (%)  |
| Mother’s education                                           | 23.058     | 28.726  | 28.726  | 11.110                 | 11.110  | 11.110  | 12.684 |
| Father’s occupation:<br>Agriculture                          | 17.863     | 19.885  | 19.885  | 4.619                  | 4.619   | 4.619   | 5.060  |
| Mother’s occupation:<br>Agriculture                          | 11.684     | 14.393  | 14.393  | 3.474                  | 3.474   | 3.474   | 3.262  |
| Household wealth                                             | 9.116      | 8.788   | 8.788   | 11.531                 | 11.531  | 11.531  | 6.980  |
| Type of residence                                            | 8.733      | 11.866  | 11.866  | 21.630                 | 21.630  | 21.630  | 21.038 |
| Skilled birth attendance                                     | 7.848      | 9.988   | 9.988   | 10.058                 | 10.058  | 10.058  | 7.107  |
| Region: Equateur                                             | 3.761      | 1.066   | 1.066   | 3.165                  | 3.165   | 3.165   | 4.165  |
| Father’s occupation:<br>Household, unskilled, not<br>working | 3.370      | 0.955   | 0.955   | -0.430                 | -0.430  | -0.430  | -0.343 |
| Region: Kinshasa                                             | 3.221      | 0.647   | 0.647   | 1.246                  | 1.246   | 1.246   | 2.054  |
| Region: Sud-Kivu                                             | 2.006      | 0.213   | 0.213   | -0.688                 | -0.688  | -0.688  | -0.434 |
| Region: Kasai Oriental                                       | 1.746      | 0.513   | 0.513   | 0.106                  | 0.106   | 0.106   | 0.096  |
| Father’s education                                           | 1.462      | 0.869   | 0.869   | 3.043                  | 3.043   | 3.043   | 3.500  |
| Region: Maniema                                              | 1.075      | 0.083   | 0.083   | 3.124                  | 3.124   | 3.124   | 4.218  |
| Region: Orientale                                            | 0.984      | 0.251   | 0.251   | 0.283                  | 0.283   | 0.283   | 0.107  |
| Region: Bas-Congo                                            | 0.678      | 0.060   | 0.060   | 0.048                  | 0.048   | 0.048   | 0.474  |
| Mother’s occupation:<br>Household, unskilled, not<br>working | 0.599      | 0.306   | 0.306   | 0.094                  | 0.094   | 0.094   | -0.068 |
| Birth order and interval:<br>5+ short interval               | 0.528      | 0.118   | 0.118   | 1.214                  | 1.214   | 1.214   | 4.518  |
| Region: Kasai Occidental                                     | 0.490      | 0.115   | 0.115   | 0.007                  | 0.007   | 0.007   | 0.102  |
| Birth order and interval:<br>5+ long interval                | 0.437      | 0.253   | 0.253   | -0.333                 | -0.333  | -0.333  | -0.040 |
| Sex of the child                                             | 0.418      | 0.454   | 0.454   | 0.068                  | 0.068   | 0.068   | -0.005 |
| Region: Katanga                                              | 0.296      | 0.067   | 0.067   | -0.155                 | -0.155  | -0.155  | -2.068 |
| Mother’s age at birth                                        | 0.261      | 0.249   | 0.249   | 0.505                  | 0.505   | 0.505   | 0.613  |
| Region: Nord-Kivu                                            | 0.208      | 0.018   | 0.018   | 0.619                  | 0.619   | 0.619   | 0.429  |
| Birth order and interval:<br>2-4 long interval               | 0.154      | 0.115   | 0.115   | 0.409                  | 0.409   | 0.409   | 0.106  |
| Birth order and interval:<br>2-4 short interval              | 0.004      | 0.001   | 0.001   | 0.458                  | 0.458   | 0.458   | 0.231  |

### 1.3 Direct methods (concentration-index trees and forests)

#### 1.3.1 Rank Dependent Concentration-Index Tree

##### The Corrected Concentration-Index Tree $CIc$

We start with a tree based on the corrected concentration index, appropriate for bounded outcomes (Erreygers, 2009). In this method, the tree shows that inequality in under-five mortality is structured primarily by maternal education, father's agricultural occupation, household wealth, skilled birth attendance, and residence in Katanga Figure 2. The lowest-mortality terminal group consists of children whose mothers had some education and whose fathers were not in agriculture, with an under-five mortality rate of about 6.5%. At the opposite end, children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga had the highest mortality rate, about 17.8%. Thus, the corrected rank-dependent tree identifies a clear socioeconomic gradient, but one that is expressed through combinations of education, occupation, household wealth, delivery care, and region rather than through wealth alone.

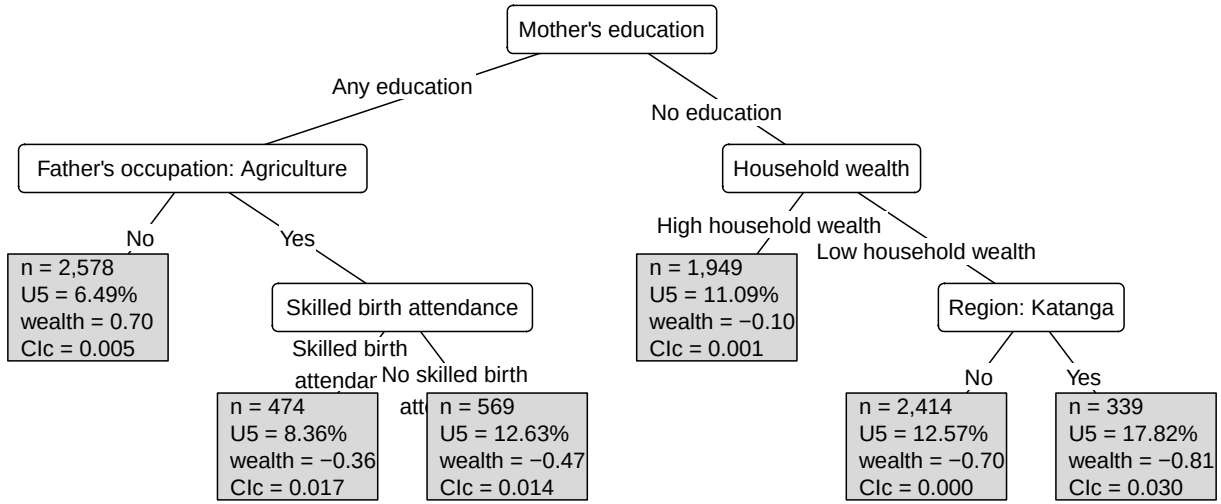


Figure 2: Concentration-index tree fitted with the  $CIc$  criterion.

##### Generalised and standard concentration-index trees $CIg$ and $CI$

The appendix also shows the trees fitted using the generalised and standard concentration indices. The generalised concentration-index tree is almost identical to the corrected concentration-index tree Figure 15. It again separates children first by maternal education and father's agricultural occupation, then by skilled birth attendance, household wealth, and residence in Katanga. The mortality pattern is therefore substantively unchanged: the lowest-risk group consists of children whose mothers had some education and whose fathers were not in agriculture, while the highest-risk group consists of children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga, with mortality around 17.8% Figure 15. The standard concentration-index tree gives a similar structure. However, it adds an extra distinction among children whose mothers had some education and whose fathers were not in agriculture, splitting them by short preceding birth interval Figure 16. Children in this otherwise advantaged group without a short birth interval have the lowest mortality, about 6.0%. In contrast, those with a short birth interval have higher mortality, about 11.7% Figure 16. Apart from this additional demographic split, the standard  $CI$  tree points to the same main sources of inequality as the

corrected and generalised trees: maternal education, father’s agricultural occupation, delivery care, household wealth, and residence in Katanga.

### 1.3.2 Level dependent Concentration-Index Tree $L$

We then fit a tree based on the level-dependent concentration index (Erreygers and Kessels, 2017) criterion,  $L$ . In this method, the selected tree shows that inequality in under-five mortality is structured primarily by rural residence, father’s agricultural occupation, household wealth, and province, especially Kinshasa, Equateur, and Katanga Figure 3. The lowest-mortality terminal group consists of urban children in Kinshasa whose fathers were not in agriculture, with an under-five mortality rate of about 5.1%. The highest-mortality terminal group consists of rural children living in low-wealth households in Katanga, with an under-five mortality rate of about 18.5%. Thus, the level-dependent tree tells a more spatial story of inequality: mortality risk is first organised by the rural-urban divide, then by occupation, wealth, and region. This shows that the level-dependent method found a different structure compared to the rank-dependent methods

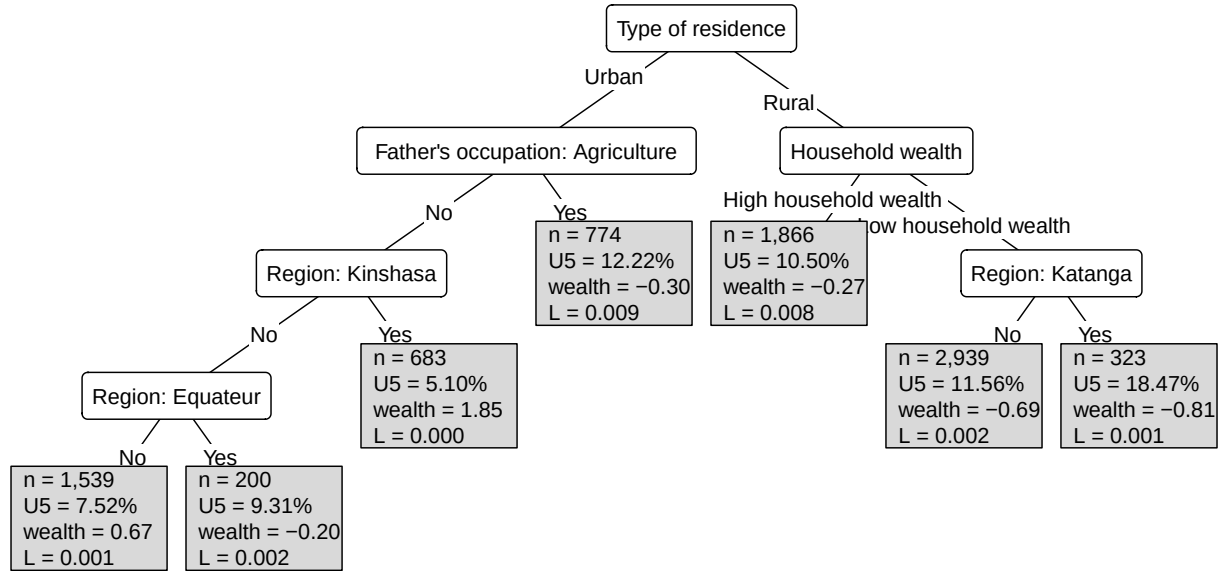


Figure 3: Level dependent concentration-index tree for under-five mortality in the DRC.

### 1.3.3 Tree validation plots

The rank-dependent trees performed well on the training data but poorly on the validation data. The CI, CIc, and CIg trees removed approximately 87 to 88% of root-node inequality during training, but had negative validation gains at the selected settings: -75.5% for CI and -73.1% for both CIc and CIg Figure 4, Table 7. In this criterion, a negative validation gain means that the sum of the terminal-node concentration-index values in the validation fold exceeded the concentration-index value at the root node. Thus, although these trees identified substantial rank-based structure in the fitted sample, the same partitions did not reduce inequality when applied to held-out folds. Unlike rank-dependent trees, the level-dependent tree maintained a positive validation gain throughout the tuning path. At the selected setting, it removed 70.5% of root-node inequality in held-out folds while retaining an average of 6.8 terminal nodes Figure 4,

Table 7. The rural-urban, occupational, wealth, and regional partition identified by the L tree was therefore more stable under the validation criterion used in this thesis.

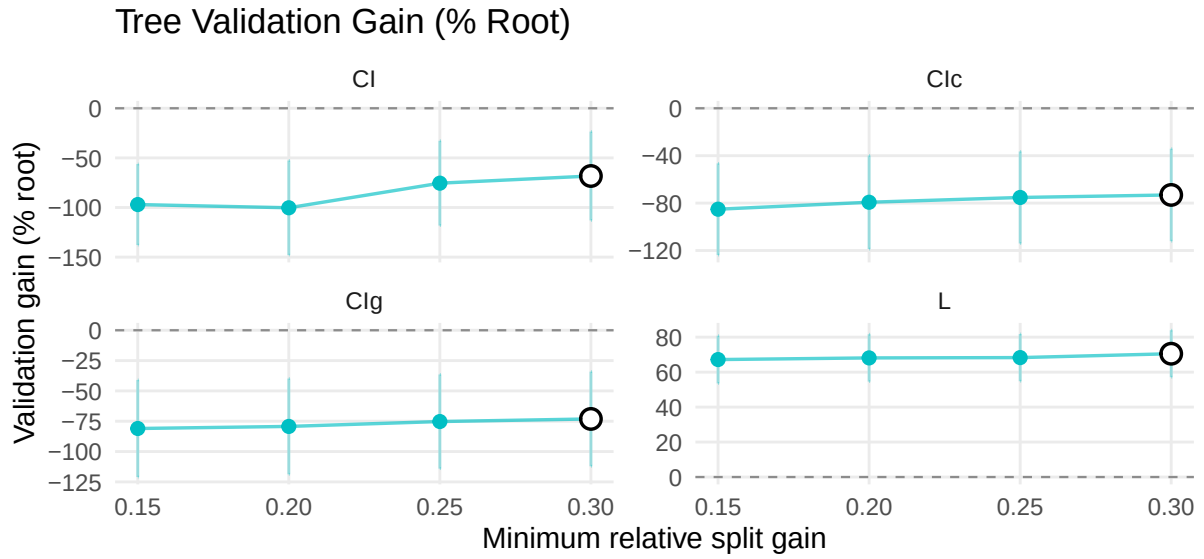


Figure 4: Tree validation gain and complexity by minimum relative split gain.

### Tree complexity plots

Tree complexity decreased as the minimum relative split gain increased, since larger thresholds allowed only splits that removed a larger share of parent-node impurity Figure 5. The selected trees remained small, with an average of 7.5 terminal nodes for CI, 7.3 for CIc and CIg, and 6.8 for L Table 8, Figure 5.

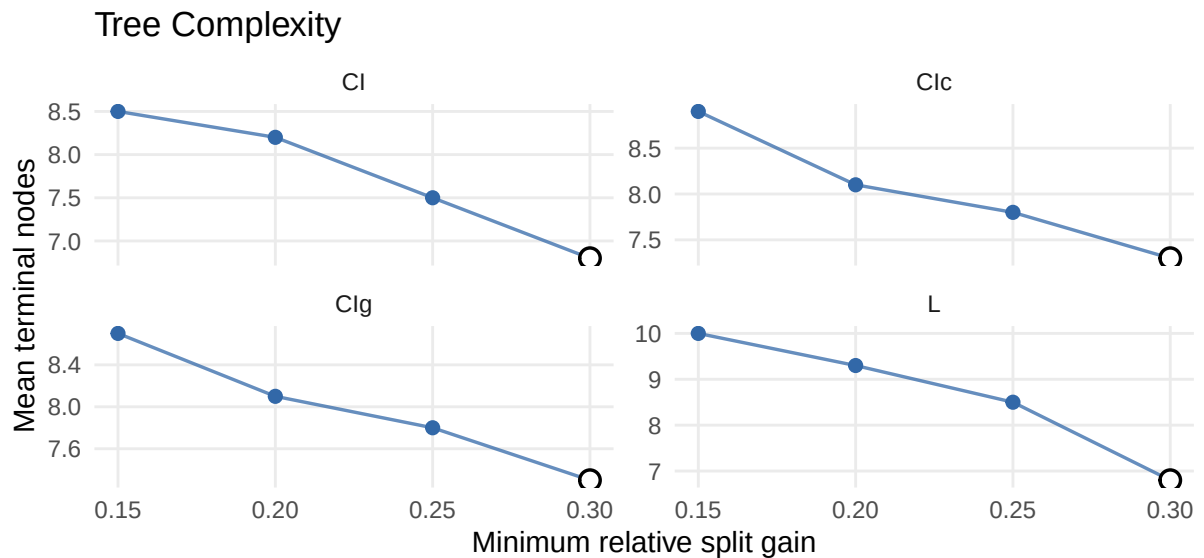


Figure 5: Tree complexity by number of terminal nodes.

### Rpart comparison

The rpart comparison tree recovered the same upper-level structure as the main CI tree (Figure 16; Figure 6). Both trees first split on maternal education, then separated children by father's agricultural occupation among those whose mothers had some education, and by household

wealth among those whose mothers had no education. The high-mortality branch was also similar, identifying children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga. The main difference was in the lower branches: the rpart comparison introduced a split by long preceding birth interval, while the main CI tree used a short preceding birth-interval split. This suggests that the main rank-dependent structure was not driven only by the custom implementation, although some lower-level refinements depended on the tree-growing and stopping rules.

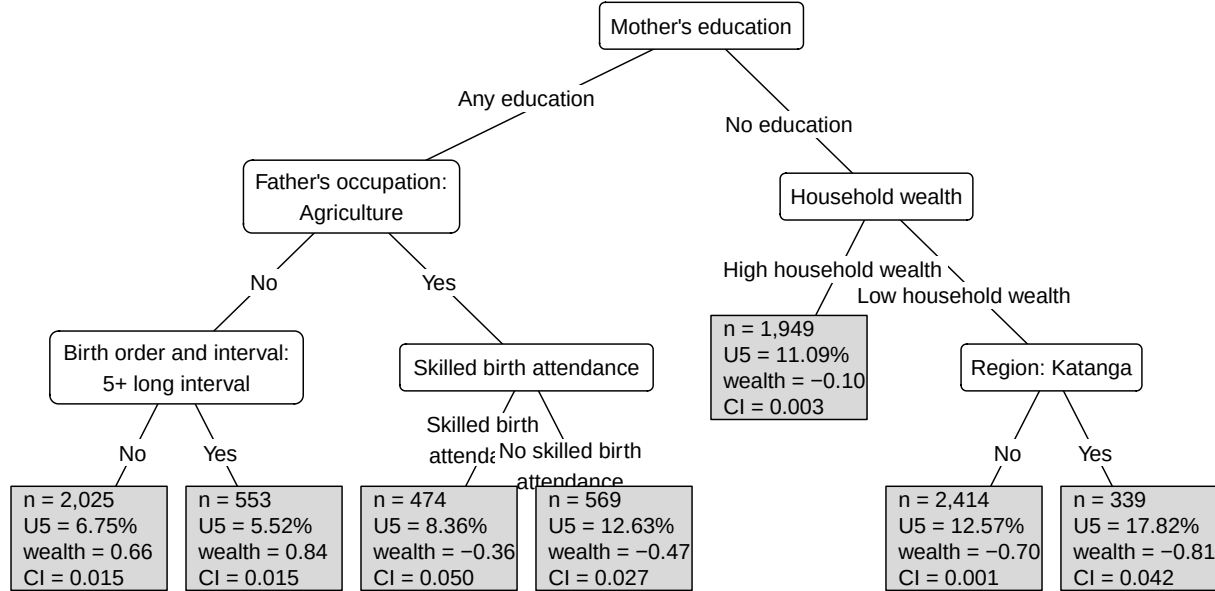


Figure 6: Rpart comparison tree using CI impurity.

### 1.3.4 Concentration index forest and surrogate tree

#### Forest validation plots

We evaluated concentration-index forests using the same validation criterion as the single trees. For the forests, the minimum relative split gain was tuned to 0.35, since larger values produced near-root-only forests with mean terminal-node counts close to 1. At that point, further validation failed to reveal a meaningful structure. For the rank-dependent criteria, the forests reduced overfitting but did not stabilise the partitions. The CI, CIc, and CIg forests removed 29.6%, 38.5%, and 38.7% of root-node inequality in training, compared with about 87 to 88% for the corresponding single trees Table 10, Table 7. Their validation gains remained negative: -25.0% for CI, -28.1% for CIc, and -34.7% for CIg Figure 7, Table 10. Thus, averaging across trees reduced the validation loss, but did not produce a reproducible rank-dependent subgroup structure. The tuning path shows the trade-off. Validation gain improved as the minimum relative split gain increased, but the component trees also became shallower Figure 7, Figure 18. At the selected setting, the mean terminal-node counts were 2.4 for CI, 2.9 for CIc, and 2.9 for CIg Table 11. The level-dependent forest remained more stable. It removed 95.1% of root-node inequality in training and 69.7% in validation Table 10. The selected L forest had an average of 4.1 terminal nodes per tree, compared with 6.8 for the selected L tree Table 11, Table 8.

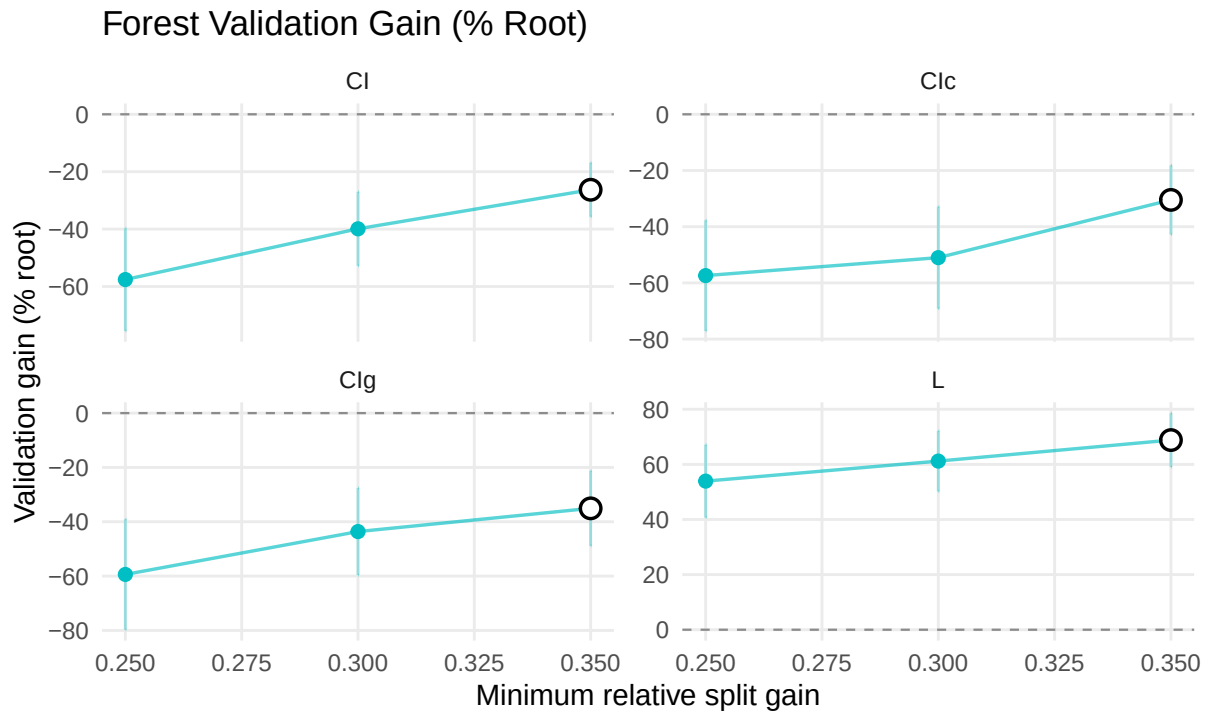


Figure 7: Forest validation gain and complexity by minimum relative split gain.

### Surrogate trees from Forest Predictions

To interpret the selected forest, we fitted a surrogate tree to summarise its predictions (Figure 8). The surrogate tree retained the main structure seen in the level-dependent tree.

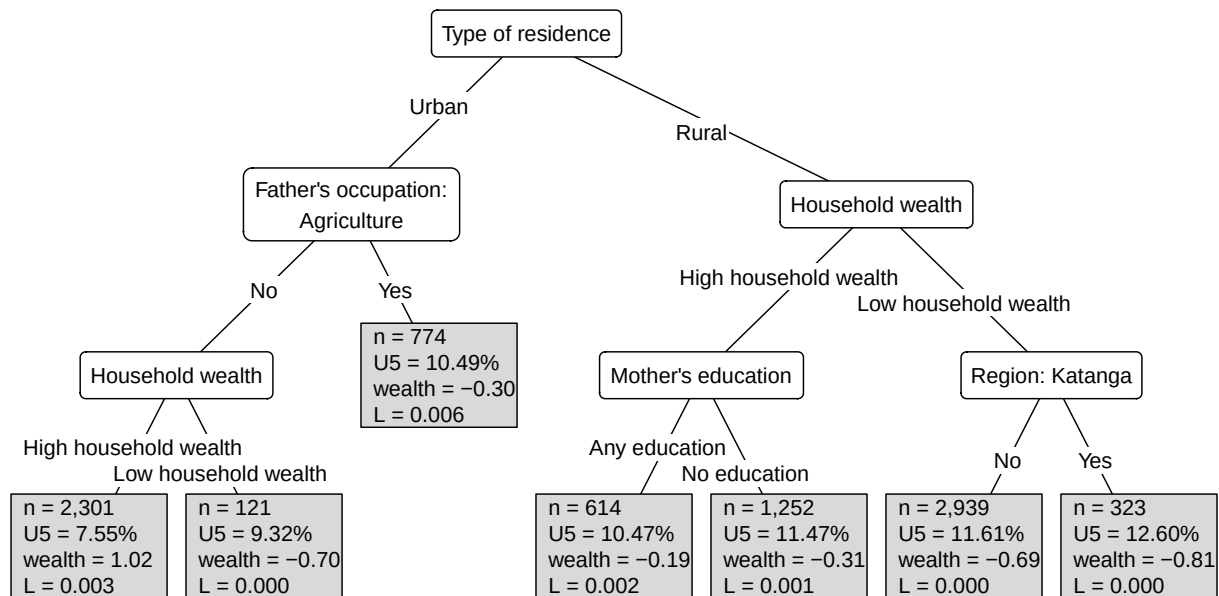


Figure 8: Surrogate tree summarising the selected concentration-index forest predictions.

The first split was type of residence, separating urban and rural children. Among urban children, the main divisions were father's agricultural occupation and household wealth. Among rural children, the main divisions were household wealth, maternal education, and residence in Katanga. The lowest terminal-node mortality was observed among urban children in the high-

wealth branch, while the highest terminal-node mortality was observed among rural, low-wealth children in Katanga. Compared with the single L tree, the surrogate gives a smoother version of the same rural-urban, socioeconomic, and regional pattern, with less extreme terminal-node mortality differences. The criterion-specific forest surrogate trees are shown in the appendix for comparison (Figure 21; Figure 22; Figure 20).

### 1.3.5 Variable importance

The variable-importance plots support the interpretation of the selected level-dependent models. In both the  $L$  tree and  $L$  forest, the type of residence gives the largest reduction in concentration-index impurity Figure 9. In the single tree, residence is followed by father’s agricultural occupation, Kinshasa, household wealth, Equateur, and Katanga, reflecting a structure organised around the rural-urban divide, occupational position, household wealth, and regional differences.

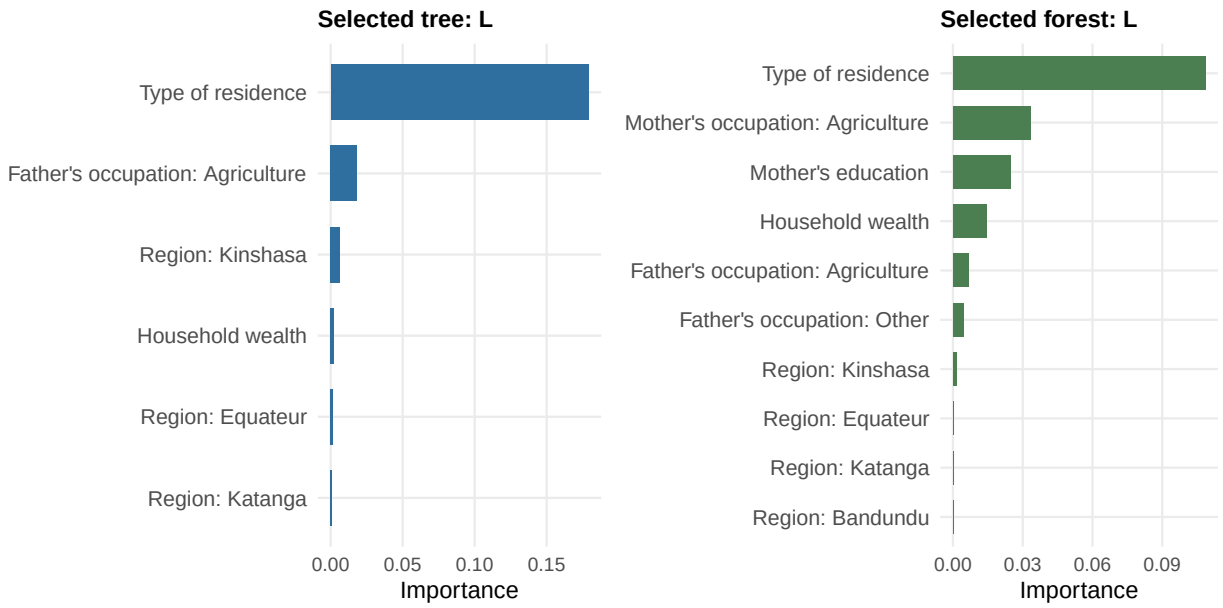


Figure 9: Variable importance for the selected concentration-index tree and forest.

The  $L$  forest gives a less concentrated ranking than the single tree. Residence is still the first variable, but the forest also gives weight to mother’s agricultural occupation, maternal education, household wealth, father’s agricultural occupation, and regional indicators. In the appendix plots, the same groups of variables appear again under the other criteria: residence, region, occupation, education, and wealth Figure 14, Figure 19.

## 1.4 Simulation Study

### 1.4.1 Null case

From (Figure 10) we see that the algorithm developed here did not find a meaningful structure and behaved as expected. This shows that the tree-growing rule does not create subgroup structure when the outcome and socioeconomic ranking are independent, since the tree uses concentration index as an impurity measure, which measures how socioeconomic ranking and outcome covary.





Figure 10: Null simulation concentration-index trees by criterion.

### 1.4.2 Poor subgroup case

In the positive-control simulation, the method (the concentration index tree with  $CIc$  as an impurity measure) recovered the planted subgroup structure. The tree first separated urban from rural observations, then split the rural group by province and maternal education (Figure 11). This structure matched the way the simulated outcome was generated: poorer observations in the planted rural, low-education, selected-province subgroup had higher outcome prevalence. The terminal nodes also followed the expected gradient, with the highest prevalence among observations in the most disadvantaged planted subgroup.

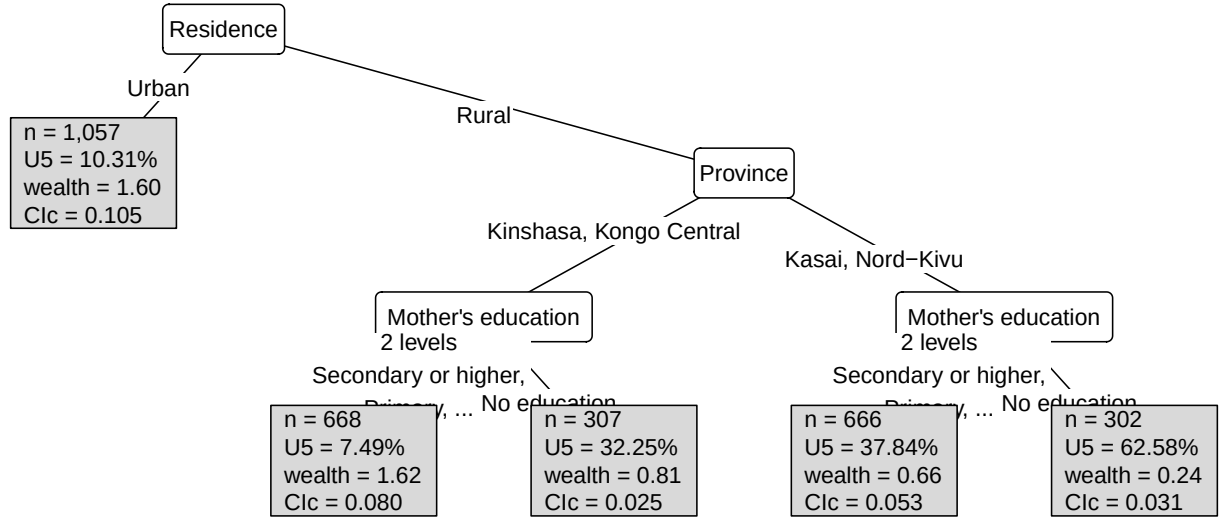


Figure 11:  $CIc$  concentration-index tree recovered from simulated data with a known subgroup-generating mechanism.

We see the same pattern across the sensitivity plots in the appendix the trees developed using  $CI$ ,  $CIg$ , and  $CIc$ , and  $L$  as impurity measures (Figure 24; Figure 25; Figure 26).

## 2 Discussion

### 2.1 Main findings

This thesis developed and applied a tree-based model for analysing socioeconomic inequality in under-five mortality using the concentration index impurity. The main finding is that inequality in the DRC data could not be explained by household wealth alone. Under-five mortality was also related to residence, maternal education, skilled birth attendance, parental occupation, and province. Overall, 10.32% of children died before age five. The children who died were more often from low-wealth households, rural households, mothers with no education, births without skilled attendance, and some provinces, especially Katanga. This is in line with earlier DRC work on geographic variation in under-five mortality, and with decomposition studies showing that wealth, education, residence, and occupation contribute to mortality inequalities in African settings (Van Malderen, Van Oyen and Speybroeck, 2013; Kandala *et al.*, 2014; Van Malderen *et al.*, 2019).

The indirect methods provided a variable view of this inequality. In the regression decomposition, mother’s education made the largest contribution, accounting for 23.1% under *CI* and 28.7% under *CIg* and *CIc*. Father’s agricultural occupation was the second-largest contributor, accounting for 17.9% under *CI* and 19.9% under *CIg* and *CIc*. These results are consistent with earlier decomposition work, which found that household wealth, parental education, occupation, and residence were important contributors to under-five mortality inequality (Van Malderen, Van Oyen and Speybroeck, 2013; Van Malderen *et al.*, 2019). The SHAP analysis (from the fitted random forest) gave a less concentrated picture: residence and household wealth each contributed about 12%, followed by mother’s occupation, father’s occupation, mother’s education, and skilled birth attendance. Shap-based decomposition provides an addition to the regression methods by allowing the use of more flexible models such as Random Forests or XGBoost.

The direct concentration-index trees added the subgroup interpretation that is less visible in the regression-based decomposition. While the indirect methods ranked variables by their contribution to inequality, the trees showed how these variables combined within the data to define groups with different inequality patterns. The rank-dependent trees based on *CI*, *CIg*, and *CIc* identified a structure involving maternal education, father’s agricultural occupation, skilled birth attendance, household wealth, and Katanga. Mortality was lower among children whose mothers had some education and whose fathers were not in agriculture, and higher among children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga. However, these partitions did not validate well. Although the rank-based methods removed about 87% to 88% of root-node inequality in training, their validation gains were negative, suggesting that the strong training structure did not transfer well to held-out folds.

### 2.2 Rank-dependent and level-dependent views of inequality

From these analyses, we see differences in the results (Figure 16; Figure 15; Figure 2) in subgroups formed by the two ways of defining the impurity measure when developing the concentration index tree. In the rank-dependent criteria *CI*, *CIg*, and *CIc*, a child’s contribution depends on the child’s position in the current node’s socioeconomic ranking. After each split, the children in the new subgroup are ranked again, so the same child can receive a different rank because the comparison group has changed. In the level-dependent criterion, re-ranking is still done but

in  $L$ , the child’s observed socioeconomic level is carried into the subgroup, so a child’s rank also reflects their wealth level.

The differences also showed up in cross-validation; the level-dependent method outperformed the rank-dependent methods. The rank-dependent trees had strong training performance but negative validation gains, while the level-dependent tree kept a positive validation gain (Figure 4; Table 7). The same pattern was seen in the forests: averaging trees reduced the validation loss for the rank-dependent criteria, but their validation gains remained below zero, whereas the  $L$  forest retained a positive validation gain (Figure 7; Table 10). The validation results show that  $CI$ ,  $CIg$ , and  $CIc$  identified detailed in-sample subgroup rules, whereas  $L$  produced a more reproducible subgroup structure in this application.

These results are also mirrored by the subgroup-decomposability argument in (Erreygers *et al.*, 2017). In their comparison of rank-dependent and level-dependent measures, they show that the level-dependent index  $L$  is subgroup-decomposable and rank-dependent measures are more difficult to decompose cleanly across population groups. Decomposability of the level-dependent concentration index shown by (Erreygers *et al.*, 2017) can also explain why  $L$  performed better as a tree impurity criterion in this application: children inherit their socioeconomic level within each subgroup, rather than being re-weighted only by a new within-node rank. Although the DHS wealth index is an asset-based score (The DHS Program, 2007) and (Erreygers and Kessels, 2017) recommends using Level dependent concentration index( $L$ ) with a wealth-ranking variable that has ratio properties. The cross-validation results show that the  $L$  concentration index impurity measure yielded a structure transferable to held-out samples using the DHS wealth score.

## 2.3 Methodological and data limitations

This thesis uses DHS survey data (The DRC Program Program, 2007). On the DHS privacy page (The DHS Program, 2026), they state that this voluntary survey data is known to have some limitations, such as non-response bias and recall bias, because deaths were recorded after the fact. We also used an older data set because we wanted to compare the new method with a data set with known results. So the results might not reflect the current status in the Democratic Republic of Congo.

The second limitation is the socioeconomic ranking variable, which is reconstructed using PCA (The DHS Program, 2007). The DHS wealth index is useful for ordering households from poorest to richest. It is also a relative score within the survey. This limitation mainly affects the level-dependent concentration index. This specifically affects  $L$ , because the level-dependent index has been shown to perform best for wealth variables with rational properties (Erreygers and Kessels, 2017).

We also used decision trees, which are known to be unstable. Small changes in the data can produce different split variables, split points, and tree structures. This instability arises from the recursive, greedy nature of tree construction: once an early split changes, all subsequent splits are fit to different subsets of the data. Since this is a concentration index tree, we tried to address this by introducing a minimum relative gain, so that we can admit splits that remove a set amount of gain, chosen via cross-validation.

## 2.4 Ethical, societal, and stakeholder relevance

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## 3 Future research

In this thesis, the level-dependent  $L$  method yielded more stable validation results than the rank-dependent criteria. This suggests that the choice of concentration index, used here as the tree impurity measure, is an important part of the method. One useful direction for future work would be to investigate whether rank-dependent concentration-index trees can be modified so that a child node retains more information about the wealth position from which its observations came, rather than relying solely on the new within-node ranking after each split. This may help reduce some of the instability seen in the rank-dependent trees. Another direction is to study the hyperparameters more systematically. Controls such as minimum node size, minimum child-node proportion, maximum depth, minimum gain, and minimum relative gain can alter the final tree structure, depending on the study’s purpose. Future work could therefore examine how these controls affect validation gain, terminal-node size, and the stability of the subgroups found by the model.

## 4 Conclusion

This thesis developed and applied a tree-based method for analysing socioeconomic inequality in under-five mortality. The method used the concentration index as the impurity measure in a recursive partitioning method. Instead of ranking determinants by their percentage contribution to inequality, the fitted tree produced subgroups that could be interpreted as decision rules. This added a subgroup view to the usual regression-based decomposition. In the DRC DHS application, inequality in under-five mortality was not explained solely by household wealth. Residence, maternal education, skilled birth attendance, parental occupation, household wealth, and province all appeared in the analysis.

The indirect methods gave the variable-level view, while the direct trees showed how these variables combined. The regression decomposition emphasised the mother’s education and the father’s agricultural occupation. The SHAP-based decomposition gave a broader pattern, with residence, household wealth, mother’s education, skilled birth attendance, and parental occupation appearing across criteria.

The concentration-index trees acted as a direct method to decomposition, which showed directly with subgroups had higher under 5 deaths concentration. The rank-dependent trees found an education-occupation-wealth pattern, with higher mortality among children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga. The  $L$  method gave a more spatial structure, beginning with residence and then splitting by occupation, wealth, and province.

In this thesis, the level-dependent  $L$  method yielded more stable validation results than the rank-dependent criteria. This suggests that the choice of concentration index, used here as the tree impurity measure, is an important part of the method. The rank-dependent trees found detailed structures in the fitted data, but these structures did not validate well. The  $L$  tree and forest retained positive validation gains, so the level-dependent result was the more stable descriptive partition in this application.

Since the level-dependent concentration index, as a measure of impurity, produced better results on held-out folds, future research should examine how to improve rank-dependent trees to retain more information from the original wealth ranking after each split.

## 5 Appendix

### 5.1 Results

#### 5.1.1 Appendix R1: Exploratory Data Outputs

Table 2: Weighted distribution of analysis variables by under-five mortality status

| Variable                    | Under-five mortality |                   | Overall            |
|-----------------------------|----------------------|-------------------|--------------------|
|                             | Alive at age 5       | Died before age 5 | Overall            |
|                             | (weighted N=7,464)   | (weighted N=859)  | (weighted N=8,323) |
| Household wealth            |                      |                   |                    |
| High household wealth       | 4,266.9 (57.2%)      | 418.6 (48.7%)     | 4,685.5 (56.3%)    |
| Low household wealth        | 3,197.4 (42.8%)      | 440.2 (51.3%)     | 3,637.6 (43.7%)    |
| Skilled.birth.attendance    |                      |                   |                    |
| Skilled.birth.attendance.1  | 3,225.3 (43.2%)      | 295.1 (34.4%)     | 3,520.5 (42.3%)    |
| No.skilled.birth.attendance | 4,239 (56.8%)        | 563.6 (65.6%)     | 4,802.6 (57.7%)    |
| Sex of the child            |                      |                   |                    |
| Female                      | 3,828.8 (51.3%)      | 390.4 (45.5%)     | 4,219.2 (50.7%)    |
| Male                        | 3,635.5 (48.7%)      | 468.4 (54.5%)     | 4,103.9 (49.3%)    |

|                                   |                 |               |                 |
|-----------------------------------|-----------------|---------------|-----------------|
| Birth order and interval          |                 |               |                 |
| First birth                       | 1,396.8 (18.7%) | 190.9 (22.2%) | 1,587.7 (19.1%) |
| 2-4 short interval                | 774.3 (10.4%)   | 110.1 (12.8%) | 884.4 (10.6%)   |
| 2-4 long interval                 | 2,603.8 (34.9%) | 216.6 (25.2%) | 2,820.3 (33.9%) |
| 5+ short interval                 | 696.6 (9.3%)    | 150.6 (17.5%) | 847.2 (10.2%)   |
| 5+ long interval                  | 1,992.8 (26.7%) | 190.7 (22.2%) | 2,183.5 (26.2%) |
| Mother's age at birth             |                 |               |                 |
| 20 or more                        | 4,256.1 (57.0%) | 472.6 (55.0%) | 4,728.8 (56.8%) |
| Less than 20                      | 3,208.1 (43.0%) | 386.2 (45.0%) | 3,594.3 (43.2%) |
| Type of residence                 |                 |               |                 |
| Urban                             | 2,932.1 (39.3%) | 263.6 (30.7%) | 3,195.7 (38.4%) |
| Rural                             | 4,532.2 (60.7%) | 595.2 (69.3%) | 5,127.4 (61.6%) |
| Mother's education                |                 |               |                 |
| Any education                     | 3,342.9 (44.8%) | 278.8 (32.5%) | 3,621.8 (43.5%) |
| No education                      | 4,121.4 (55.2%) | 579.9 (67.5%) | 4,701.3 (56.5%) |
| Father's education                |                 |               |                 |
| Any education                     | 5,490 (73.6%)   | 589.1 (68.6%) | 6,079.2 (73.0%) |
| No education                      | 1,974.3 (26.4%) | 269.7 (31.4%) | 2,243.9 (27.0%) |
| Mother's occupation               |                 |               |                 |
| Other                             | 1,610.2 (21.6%) | 135.1 (15.7%) | 1,745.4 (21.0%) |
| Household, unskilled, not working | 1,757.2 (23.5%) | 172.1 (20.0%) | 1,929.4 (23.2%) |
| Agriculture                       | 4,096.8 (54.9%) | 551.6 (64.2%) | 4,648.4 (55.8%) |
| Father's occupation               |                 |               |                 |
| Other                             | 2,810.3 (37.6%) | 242.1 (28.2%) | 3,052.4 (36.7%) |
| Household, unskilled, not working | 961.3 (12.9%)   | 108.5 (12.6%) | 1,069.8 (12.9%) |
| Agriculture                       | 3,692.7 (49.5%) | 508.1 (59.2%) | 4,200.9 (50.5%) |

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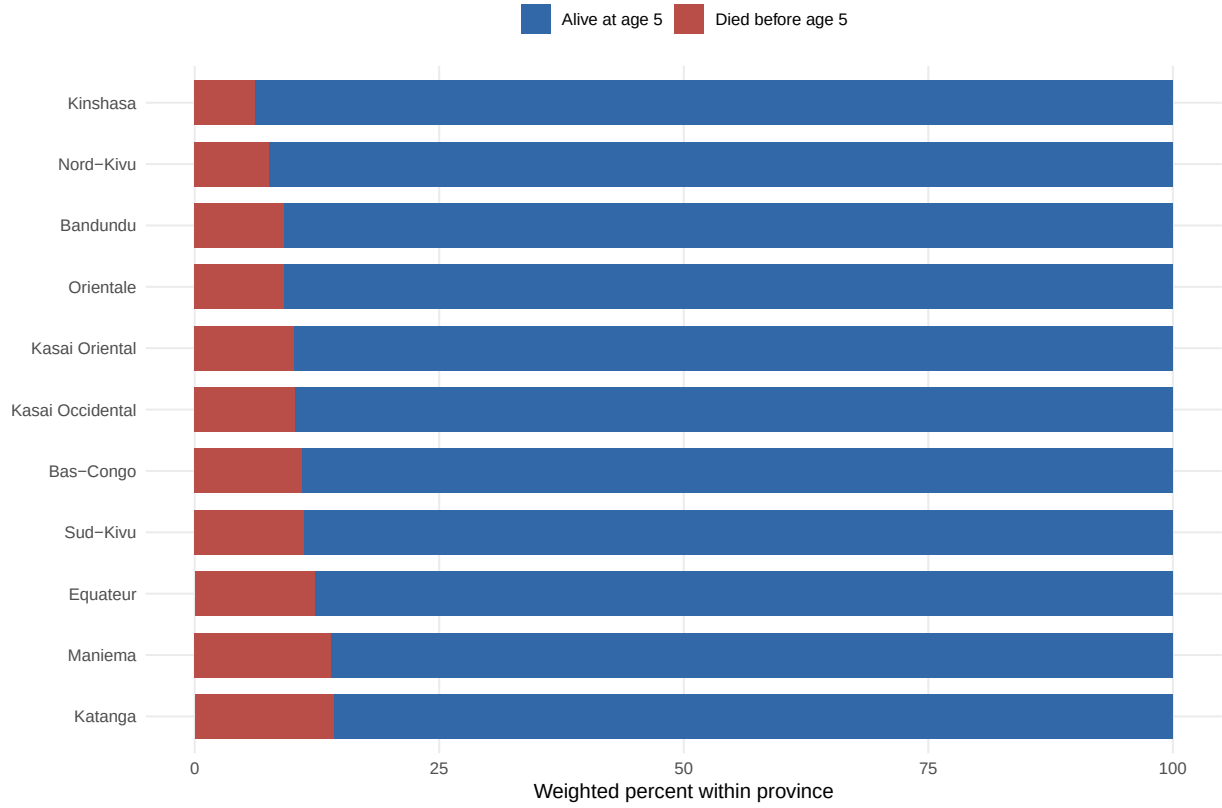


Figure 12: Weighted province distribution by under-five mortality status.

### 5.1.2 Appendix R2: Regression and Predictive-Forest Details

#### Regression model summary

Table 3: Survey-weighted regression model summary

| Term                                            | Estimate | Std. error | Statistic | p-value | Conf. low | Conf. high |
|-------------------------------------------------|----------|------------|-----------|---------|-----------|------------|
| Intercept                                       | -2.908   | 0.275      | -10.563   | <0.001  | -3.448    | -2.369     |
| Household wealth                                | 0.068    | 0.125      | 0.544     | 0.5865  | -0.177    | 0.314      |
| Skilled birth attendance                        | 0.198    | 0.111      | 1.783     | 0.0746  | -0.020    | 0.416      |
| Sex of the child                                | 0.242    | 0.105      | 2.294     | 0.0218  | 0.035     | 0.449      |
| Birth order and interval:<br>2-4 short interval | -0.004   | 0.186      | -0.020    | 0.9838  | -0.367    | 0.360      |
| Birth order and interval:<br>2-4 long interval  | -0.509   | 0.169      | -3.013    | 0.0026  | -0.840    | -0.178     |
| Birth order and interval:<br>5+ short interval  | 0.420    | 0.210      | 1.995     | 0.0461  | 0.007     | 0.832      |
| Birth order and interval:<br>5+ long interval   | -0.396   | 0.211      | -1.882    | 0.0599  | -0.809    | 0.017      |
| Mother's age at birth                           | 0.033    | 0.143      | 0.229     | 0.8186  | -0.247    | 0.312      |
| Type of residence                               | 0.090    | 0.154      | 0.584     | 0.5591  | -0.212    | 0.392      |
| Mother's education                              | 0.336    | 0.132      | 2.555     | 0.0106  | 0.078     | 0.594      |
| Father's education                              | -0.036   | 0.128      | -0.280    | 0.7798  | -0.286    | 0.214      |

|                                                              |        |       |        |        |        |       |
|--------------------------------------------------------------|--------|-------|--------|--------|--------|-------|
| Mother's occupation:<br>Household, unskilled, not<br>working | -0.018 | 0.155 | -0.114 | 0.9093 | -0.322 | 0.287 |
| Mother's occupation:<br>Agriculture                          | 0.140  | 0.163 | 0.861  | 0.3895 | -0.180 | 0.460 |
| Father's occupation:<br>Household, unskilled, not<br>working | 0.235  | 0.162 | 1.450  | 0.1472 | -0.083 | 0.554 |
| Father's occupation:<br>Agriculture                          | 0.225  | 0.149 | 1.509  | 0.1314 | -0.067 | 0.517 |
| Region: Bas-Congo                                            | 0.258  | 0.243 | 1.062  | 0.2883 | -0.218 | 0.734 |
| Region: Equateur                                             | 0.250  | 0.238 | 1.051  | 0.2934 | -0.216 | 0.715 |
| Region: Kasai Occidental                                     | 0.105  | 0.228 | 0.460  | 0.6452 | -0.342 | 0.553 |
| Region: Kasai Oriental                                       | 0.153  | 0.240 | 0.635  | 0.5251 | -0.318 | 0.624 |
| Region: Katanga                                              | 0.515  | 0.228 | 2.258  | 0.0240 | 0.068  | 0.962 |
| Region: Kinshasa                                             | 0.081  | 0.262 | 0.310  | 0.7564 | -0.433 | 0.596 |
| Region: Maniema                                              | 0.489  | 0.224 | 2.178  | 0.0294 | 0.049  | 0.929 |
| Region: Nord-Kivu                                            | -0.252 | 0.294 | -0.857 | 0.3915 | -0.827 | 0.324 |
| Region: Orientale                                            | -0.084 | 0.265 | -0.316 | 0.7522 | -0.603 | 0.435 |
| Region: Sud-Kivu                                             | 0.300  | 0.270 | 1.111  | 0.2667 | -0.229 | 0.829 |

## Predictive ranger tuning results

Table 4: Weighted class distribution before and after ranger undersampling

| Sample               | Outcome           | Rows | Weighted N | Weighted percent |
|----------------------|-------------------|------|------------|------------------|
| Before undersampling | Alive at age 5    | 7424 | 7464.30    | 89.68            |
| Before undersampling | Died before age 5 | 872  | 858.80     | 10.32            |
| After undersampling  | Alive at age 5    | 1274 | 1289.74    | 60.03            |
| After undersampling  | Died before age 5 | 872  | 858.80     | 39.97            |

Table 5: Cross-validation results for the predictive ranger model

| Metric   | mtry | min_n | Mean   | Std. error | Folds | Rank |
|----------|------|-------|--------|------------|-------|------|
| accuracy | 22   | 500   | 0.6118 | 0.0083     | 10    | 1    |
| accuracy | 10   | 200   | 0.6108 | 0.0090     | 10    | 2    |
| accuracy | 16   | 500   | 0.6104 | 0.0096     | 10    | 3    |
| accuracy | 29   | 200   | 0.6090 | 0.0106     | 10    | 4    |
| accuracy | 10   | 500   | 0.6080 | 0.0081     | 10    | 5    |
| accuracy | 22   | 200   | 0.6080 | 0.0098     | 10    | 6    |
| accuracy | 16   | 800   | 0.6076 | 0.0042     | 10    | 7    |
| accuracy | 29   | 800   | 0.6076 | 0.0050     | 10    | 8    |
| accuracy | 10   | 800   | 0.6072 | 0.0032     | 10    | 9    |



|             |    |     |        |        |    |    |
|-------------|----|-----|--------|--------|----|----|
| accuracy    | 22 | 800 | 0.6071 | 0.0051 | 10 | 10 |
| accuracy    | 16 | 200 | 0.6066 | 0.0084 | 10 | 11 |
| accuracy    | 29 | 500 | 0.6053 | 0.0087 | 10 | 12 |
| f_meas      | 29 | 200 | 0.3888 | 0.0175 | 10 | 1  |
| f_meas      | 22 | 200 | 0.3753 | 0.0185 | 10 | 2  |
| f_meas      | 16 | 200 | 0.3736 | 0.0146 | 10 | 3  |
| f_meas      | 10 | 200 | 0.3655 | 0.0155 | 10 | 4  |
| f_meas      | 22 | 500 | 0.3459 | 0.0158 | 10 | 5  |
| f_meas      | 29 | 500 | 0.3425 | 0.0152 | 10 | 6  |
| f_meas      | 16 | 500 | 0.3222 | 0.0176 | 10 | 7  |
| f_meas      | 10 | 500 | 0.2867 | 0.0200 | 10 | 8  |
| f_meas      | 29 | 800 | 0.2367 | 0.0157 | 10 | 9  |
| f_meas      | 22 | 800 | 0.2266 | 0.0194 | 10 | 10 |
| f_meas      | 16 | 800 | 0.2068 | 0.0144 | 10 | 11 |
| f_meas      | 10 | 800 | 0.1574 | 0.0167 | 10 | 12 |
| roc_auc     | 10 | 500 | 0.6158 | 0.0097 | 10 | 1  |
| roc_auc     | 10 | 200 | 0.6155 | 0.0111 | 10 | 2  |
| roc_auc     | 16 | 500 | 0.6153 | 0.0102 | 10 | 3  |
| roc_auc     | 22 | 500 | 0.6146 | 0.0100 | 10 | 4  |
| roc_auc     | 29 | 500 | 0.6141 | 0.0099 | 10 | 5  |
| roc_auc     | 16 | 200 | 0.6140 | 0.0104 | 10 | 6  |
| roc_auc     | 22 | 200 | 0.6132 | 0.0105 | 10 | 7  |
| roc_auc     | 29 | 200 | 0.6126 | 0.0103 | 10 | 8  |
| roc_auc     | 10 | 800 | 0.6100 | 0.0101 | 10 | 9  |
| roc_auc     | 29 | 800 | 0.6084 | 0.0100 | 10 | 10 |
| roc_auc     | 16 | 800 | 0.6082 | 0.0102 | 10 | 11 |
| roc_auc     | 22 | 800 | 0.6082 | 0.0104 | 10 | 12 |
| sensitivity | 29 | 200 | 0.3072 | 0.0162 | 10 | 1  |
| sensitivity | 22 | 200 | 0.2923 | 0.0188 | 10 | 2  |
| sensitivity | 16 | 200 | 0.2900 | 0.0143 | 10 | 3  |
| sensitivity | 10 | 200 | 0.2774 | 0.0152 | 10 | 4  |
| sensitivity | 22 | 500 | 0.2546 | 0.0159 | 10 | 5  |
| sensitivity | 29 | 500 | 0.2546 | 0.0149 | 10 | 6  |
| sensitivity | 16 | 500 | 0.2293 | 0.0152 | 10 | 7  |
| sensitivity | 10 | 500 | 0.1960 | 0.0161 | 10 | 8  |
| sensitivity | 29 | 800 | 0.1514 | 0.0123 | 10 | 9  |
| sensitivity | 22 | 800 | 0.1445 | 0.0149 | 10 | 10 |
| sensitivity | 16 | 800 | 0.1273 | 0.0103 | 10 | 11 |
| sensitivity | 10 | 800 | 0.0917 | 0.0109 | 10 | 12 |
| spec        | 10 | 800 | 0.9600 | 0.0056 | 10 | 1  |
| spec        | 16 | 800 | 0.9364 | 0.0088 | 10 | 2  |
| spec        | 22 | 800 | 0.9238 | 0.0115 | 10 | 3  |
| spec        | 29 | 800 | 0.9199 | 0.0089 | 10 | 4  |
| spec        | 10 | 500 | 0.8900 | 0.0102 | 10 | 5  |
| spec        | 16 | 500 | 0.8712 | 0.0131 | 10 | 6  |

|      |    |     |        |        |    |    |
|------|----|-----|--------|--------|----|----|
| spec | 22 | 500 | 0.8563 | 0.0142 | 10 | 7  |
| spec | 29 | 500 | 0.8453 | 0.0147 | 10 | 8  |
| spec | 10 | 200 | 0.8391 | 0.0148 | 10 | 9  |
| spec | 22 | 200 | 0.8241 | 0.0151 | 10 | 10 |
| spec | 16 | 200 | 0.8234 | 0.0131 | 10 | 11 |
| spec | 29 | 200 | 0.8155 | 0.0130 | 10 | 12 |

**Predictive ranger SHAP summary plot**

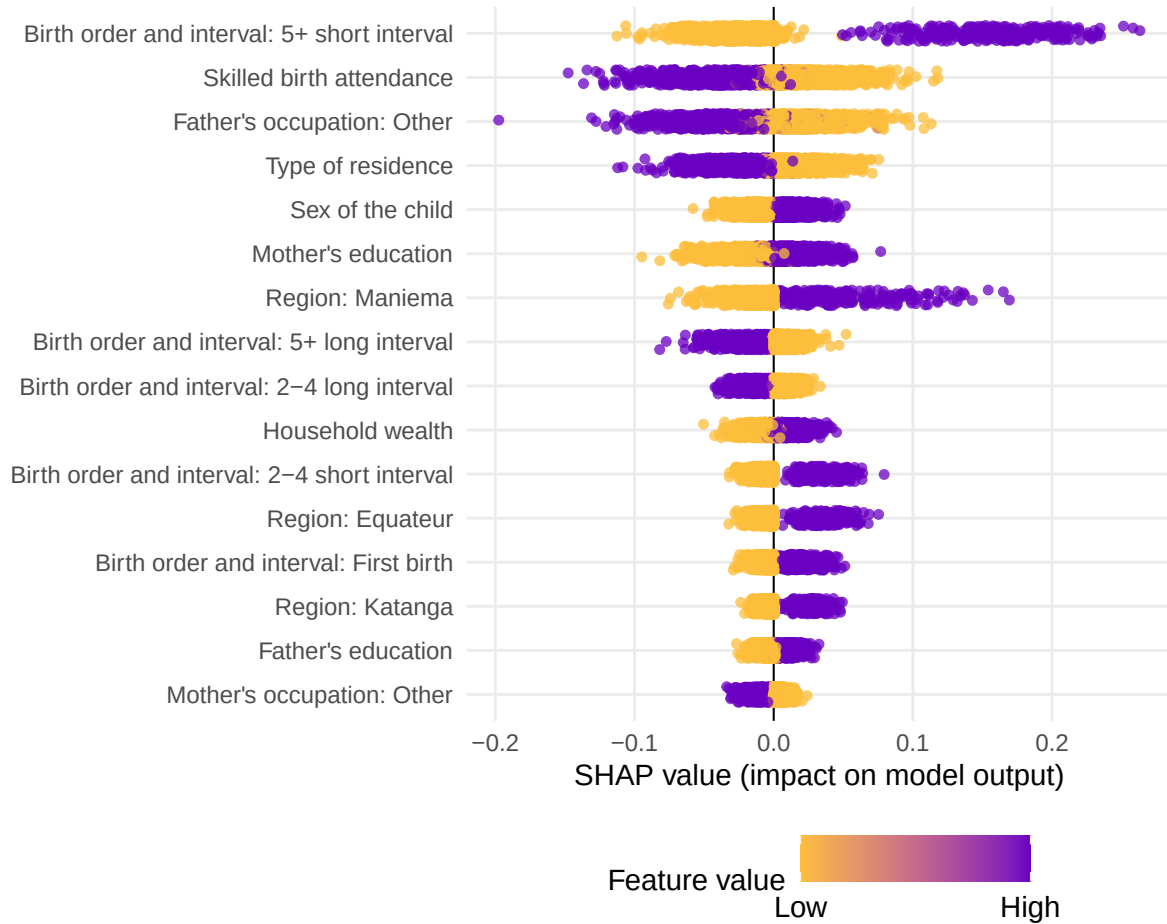


Figure 13: SHAP summary plot for the predictive ranger model.

### 5.1.3 Appendix R3: Concentration-Index Tree Details

#### Tree tuning and selection details

Table 6: Best tree hyperparameters selected by validation gain within each impurity criterion

| type | minsplit | minbucket | minprob | maxdepth | min_gain | min_relative_gain |
|------|----------|-----------|---------|----------|----------|-------------------|
| CI   | 500      | 100       | 0.01    | 10       | 1e-04    | 0.3               |
| CIg  | 500      | 100       | 0.01    | 10       | 1e-04    | 0.3               |
| CIc  | 500      | 100       | 0.01    | 10       | 1e-04    | 0.3               |

|   |     |     |      |    |       |     |
|---|-----|-----|------|----|-------|-----|
| L | 500 | 100 | 0.01 | 10 | 1e-04 | 0.3 |
|---|-----|-----|------|----|-------|-----|

### Tree fit and selection summaries

Table 7: Tree summary for the selected settings

| crit | metric                   | estimate | std_error |
|------|--------------------------|----------|-----------|
| CI   | Root impurity            | 0.1191   | NA        |
| CI   | Training gain            | 0.1003   | 0.0016    |
| CI   | Training gain (% root)   | 87.0688  | 1.5175    |
| CI   | Validation gain          | -0.0144  | 0.0236    |
| CI   | Validation gain (% root) | -68.2424 | 44.7524   |
| CIc  | Root impurity            | 0.0483   | NA        |
| CIc  | Training gain            | 0.0417   | 0.0008    |
| CIc  | Training gain (% root)   | 87.5995  | 1.4760    |
| CIc  | Validation gain          | -0.0107  | 0.0134    |
| CIc  | Validation gain (% root) | -73.1067 | 39.0032   |
| CIg  | Root impurity            | 0.0121   | NA        |
| CIg  | Training gain            | 0.0104   | 0.0002    |
| CIg  | Training gain (% root)   | 87.5995  | 1.4760    |
| CIg  | Validation gain          | -0.0027  | 0.0034    |
| CIg  | Validation gain (% root) | -73.1067 | 39.0032   |
| L    | Root impurity            | 0.3128   | NA        |
| L    | Training gain            | 0.1906   | 0.0061    |
| L    | Training gain (% root)   | 98.1536  | 0.2271    |
| L    | Validation gain          | 0.2861   | 0.1077    |
| L    | Validation gain (% root) | 70.5254  | 13.4254   |

### Tree selection metrics

Table 8: Selection metrics for fitted concentration-index trees

| Criterion | Root objective | Train gain | Train gain/root | Validation gain | Validation gain/root | Terminal nodes |
|-----------|----------------|------------|-----------------|-----------------|----------------------|----------------|
| CI        | 0.1191         | 0.1003     | 0.8707          | -0.0144         | -0.6824              | 6.8            |
| CIc       | 0.0483         | 0.0417     | 0.8760          | -0.0107         | -0.7311              | 7.3            |
| CIg       | 0.0121         | 0.0104     | 0.8760          | -0.0027         | -0.7311              | 7.3            |
| L         | 0.3128         | 0.1906     | 0.9815          | 0.2861          | 0.7053               | 6.8            |

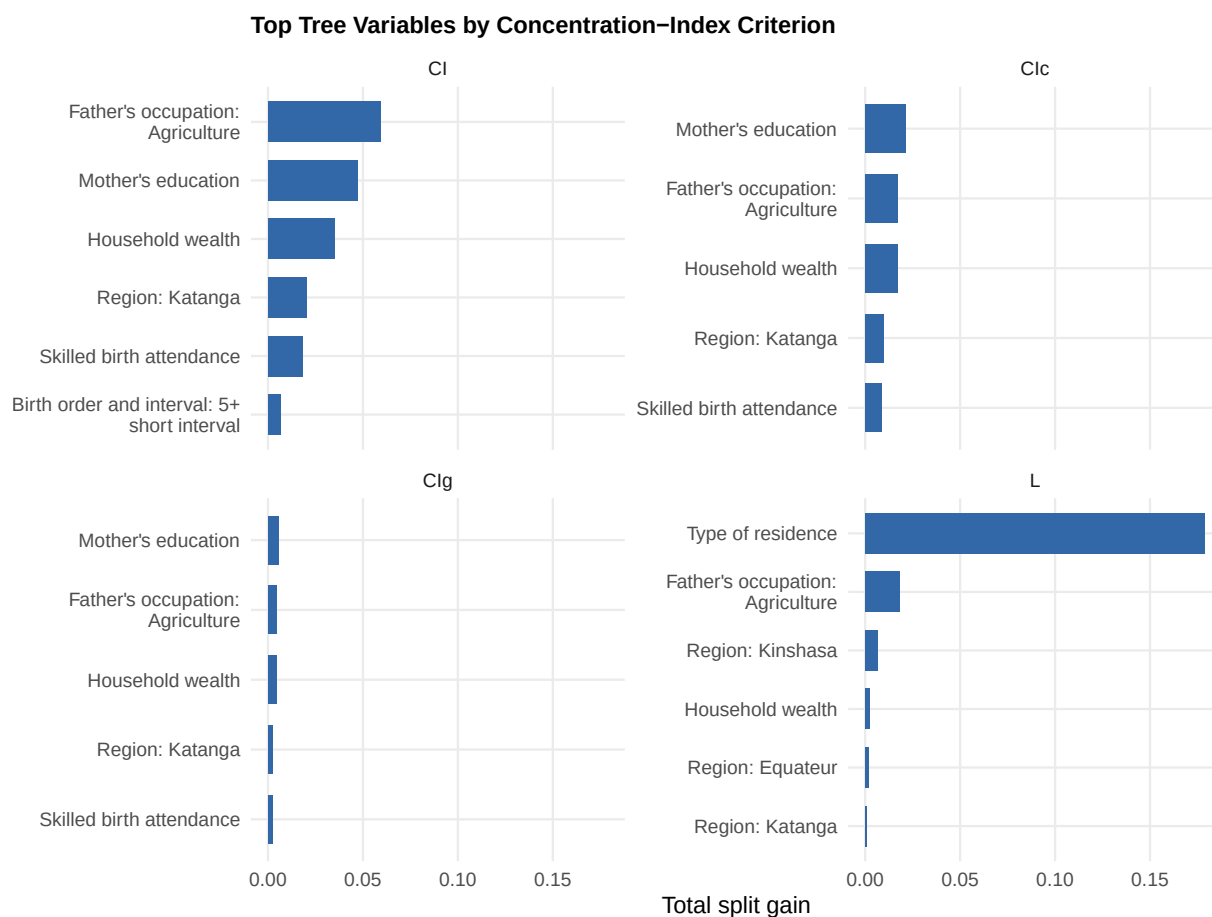


Figure 14: Top tree variables by concentration-index criterion.

#### 5.1.4 Additional Concentration-Index Tree Plots

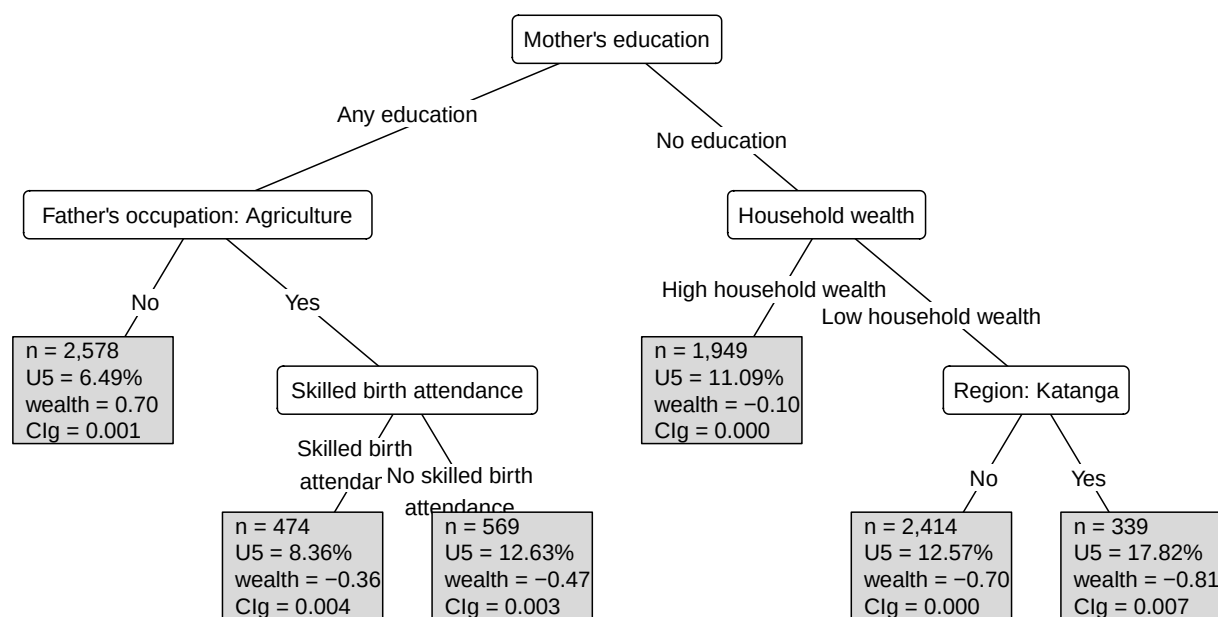


Figure 15: Concentration-index tree fitted with the CIg criterion.

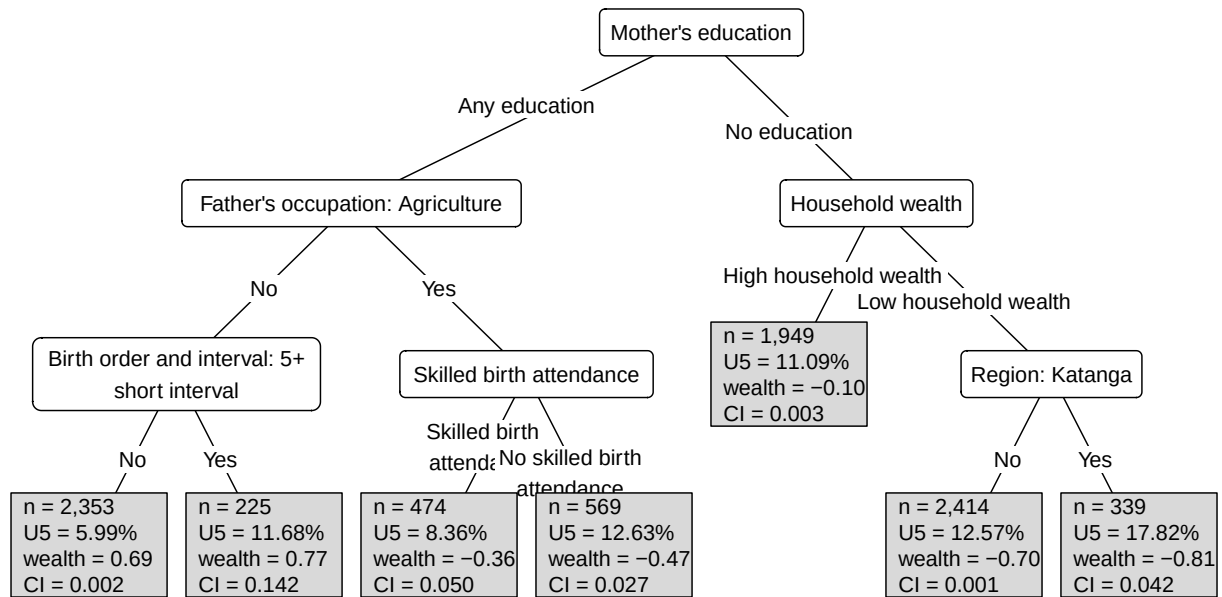


Figure 16: Concentration-index tree fitted with the CI criterion.

### 5.1.5 Appendix R4: Rpart Comparison Tree Details

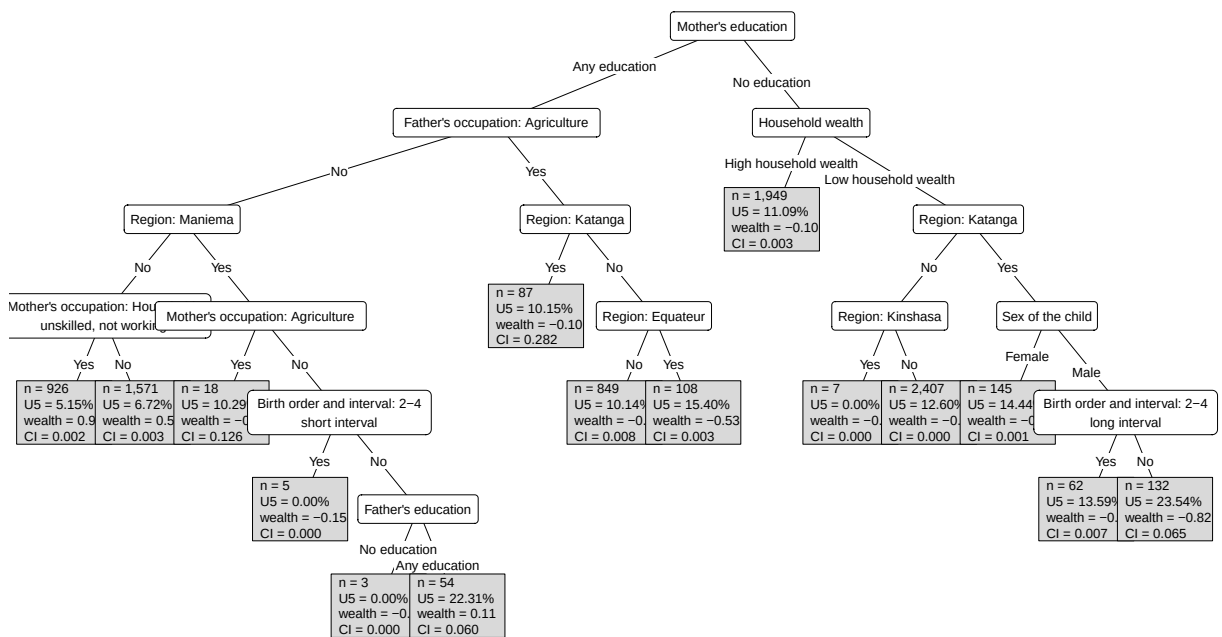


Figure 17: Lecturer rpart comparison tree using absolute CI impurity.

### 5.1.6 Appendix 5: Concentration-Index Forest Details

#### Forest tuning and selection details

Table 9: Best forest hyperparameters selected by validation gain within each impurity criterion

| type | ntree | mtry | minsplit | minbucket | minprob | maxdepth | min_gain | min_relative_gain |
|------|-------|------|----------|-----------|---------|----------|----------|-------------------|
|------|-------|------|----------|-----------|---------|----------|----------|-------------------|

|     |    |    |     |     |      |    |   |     |
|-----|----|----|-----|-----|------|----|---|-----|
| CI  | 50 | 15 | 500 | 100 | 0.01 | 15 | 0 | 0.3 |
| CIc | 50 | 15 | 500 | 100 | 0.01 | 15 | 0 | 0.3 |
| CIg | 50 | 15 | 500 | 100 | 0.01 | 15 | 0 | 0.3 |
| L   | 50 | 15 | 500 | 100 | 0.01 | 15 | 0 | 0.3 |

## Forest fit and selection summaries

Table 10: Forest summary for the selected settings

| crit | metric                   | estimate | std_error |
|------|--------------------------|----------|-----------|
| CI   | Root impurity            | 0.1191   | NA        |
| CI   | Training gain            | 0.0337   | 0.0029    |
| CI   | Training gain (% root)   | 29.5891  | 2.8849    |
| CI   | Validation gain          | -0.0075  | 0.0107    |
| CI   | Validation gain (% root) | -24.9836 | 13.0698   |
| CIc  | Root impurity            | 0.0483   | NA        |
| CIc  | Training gain            | 0.0182   | 0.0012    |
| CIc  | Training gain (% root)   | 38.5250  | 2.8199    |
| CIc  | Validation gain          | -0.0025  | 0.0054    |
| CIc  | Validation gain (% root) | -28.0573 | 16.5269   |
| CIg  | Root impurity            | 0.0121   | NA        |
| CIg  | Training gain            | 0.0046   | 0.0003    |
| CIg  | Training gain (% root)   | 38.6693  | 2.1250    |
| CIg  | Validation gain          | -0.0009  | 0.0015    |
| CIg  | Validation gain (% root) | -34.6842 | 19.5160   |
| L    | Root impurity            | 0.3128   | NA        |
| L    | Training gain            | 0.1847   | 0.0062    |
| L    | Training gain (% root)   | 95.0817  | 0.1536    |
| L    | Validation gain          | 0.2831   | 0.1064    |
| L    | Validation gain (% root) | 69.6578  | 13.3968   |

## Forest selection metrics

Table 11: Selection metrics for fitted concentration-index forests

| Criterion | Root objective | Train gain | Train gain/root | Validation gain | Validation gain/root | Mean terminal nodes |
|-----------|----------------|------------|-----------------|-----------------|----------------------|---------------------|
| CI        | 0.1191         | 0.0337     | 0.2959          | -0.0075         | -0.2498              | 2.442               |
| CIc       | 0.0483         | 0.0182     | 0.3852          | -0.0025         | -0.2806              | 2.864               |
| CIg       | 0.0121         | 0.0046     | 0.3867          | -0.0009         | -0.3468              | 2.930               |
| L         | 0.3128         | 0.1847     | 0.9508          | 0.2831          | 0.6966               | 4.070               |

## Forest complexity plots

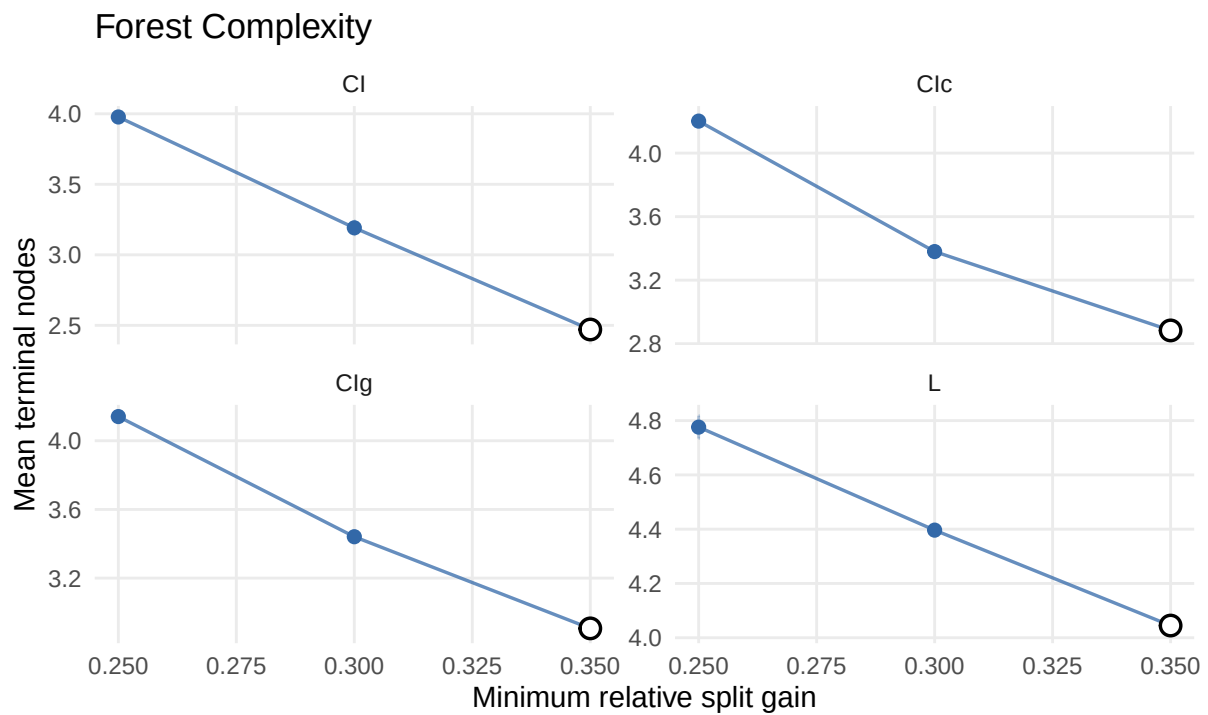


Figure 18: Forest complexity by number of terminal nodes.

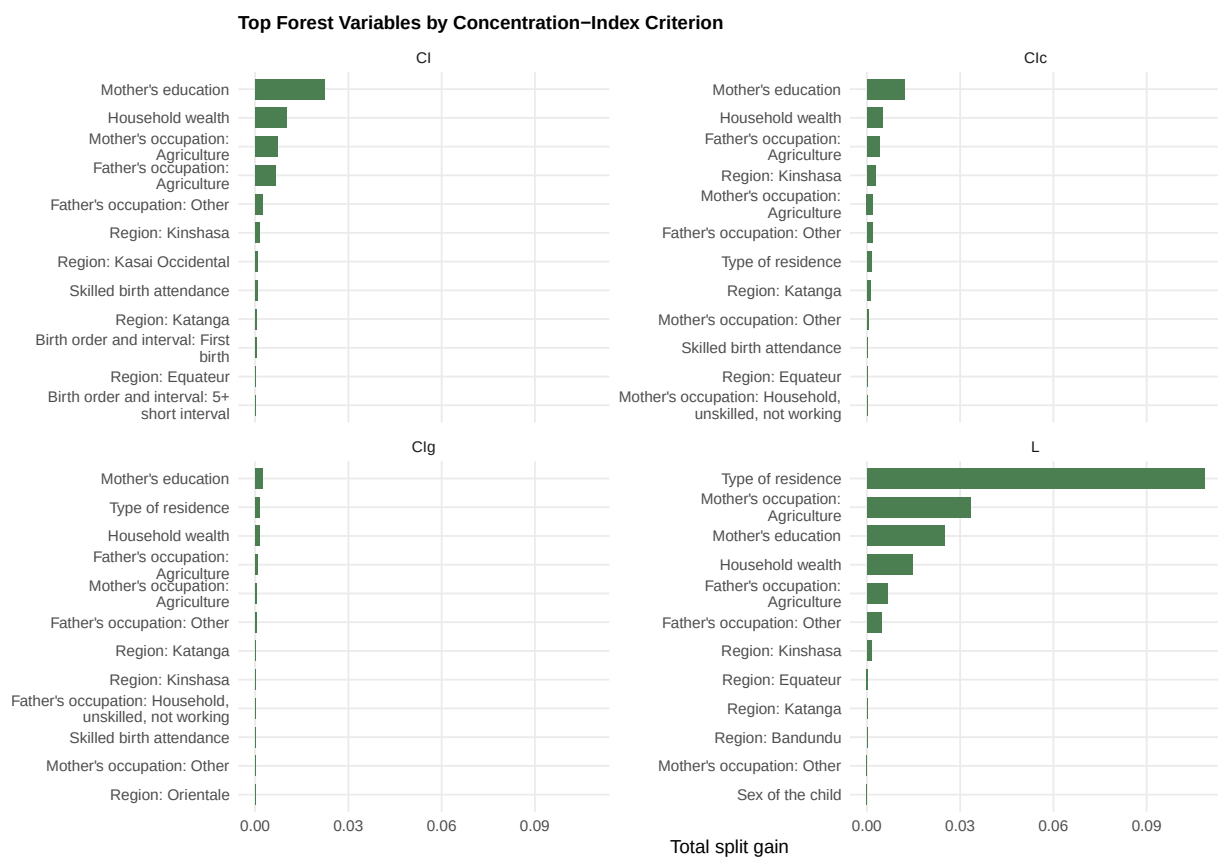


Figure 19: Top forest variables by concentration-index criterion.

## Forest Surrogate Trees plots

The selected forest surrogate appears in the results section. The remaining criterion-specific forest surrogates are retained here for comparison.

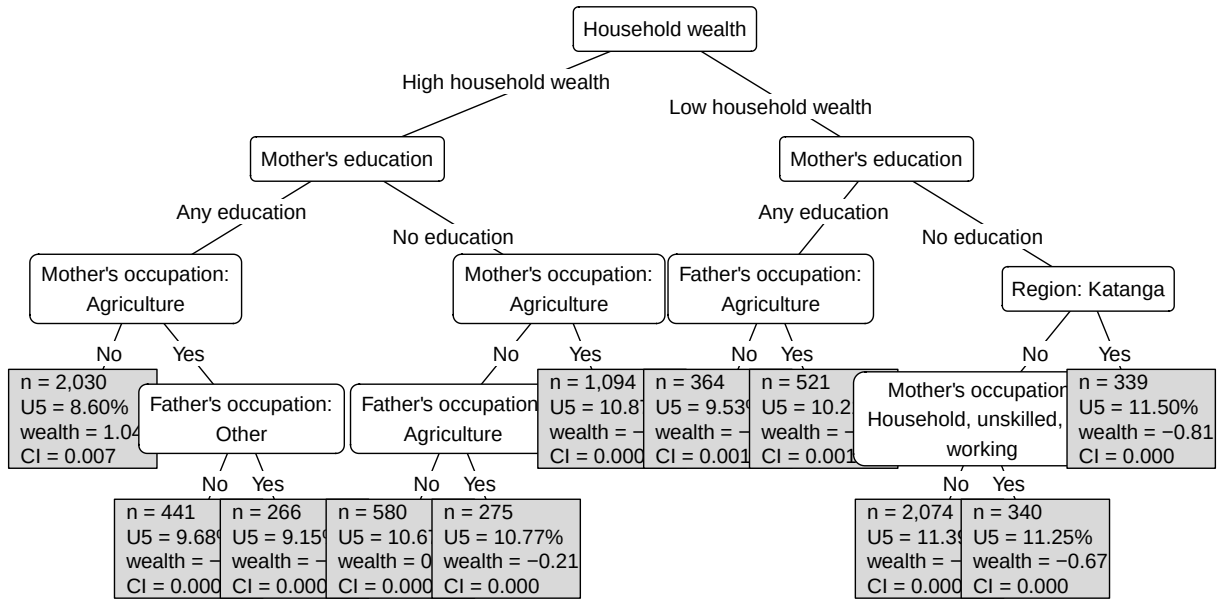


Figure 20: Surrogate tree for the CI concentration-index forest.

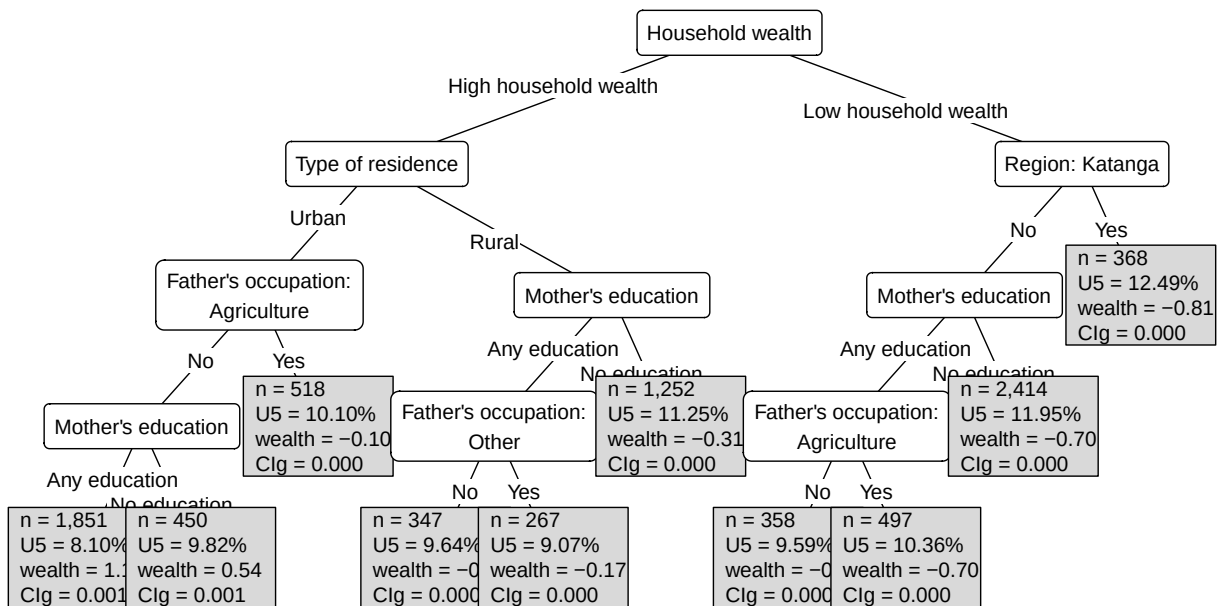


Figure 21: Surrogate tree for the CIg concentration-index forest.



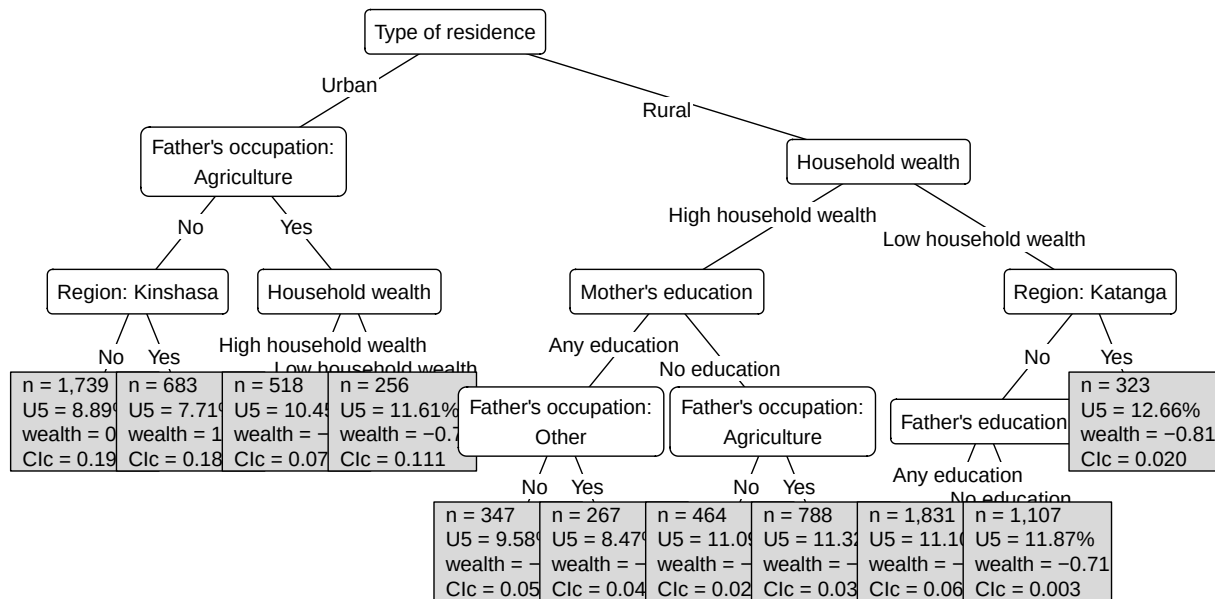


Figure 22: Surrogate tree for the Clc concentration-index forest.

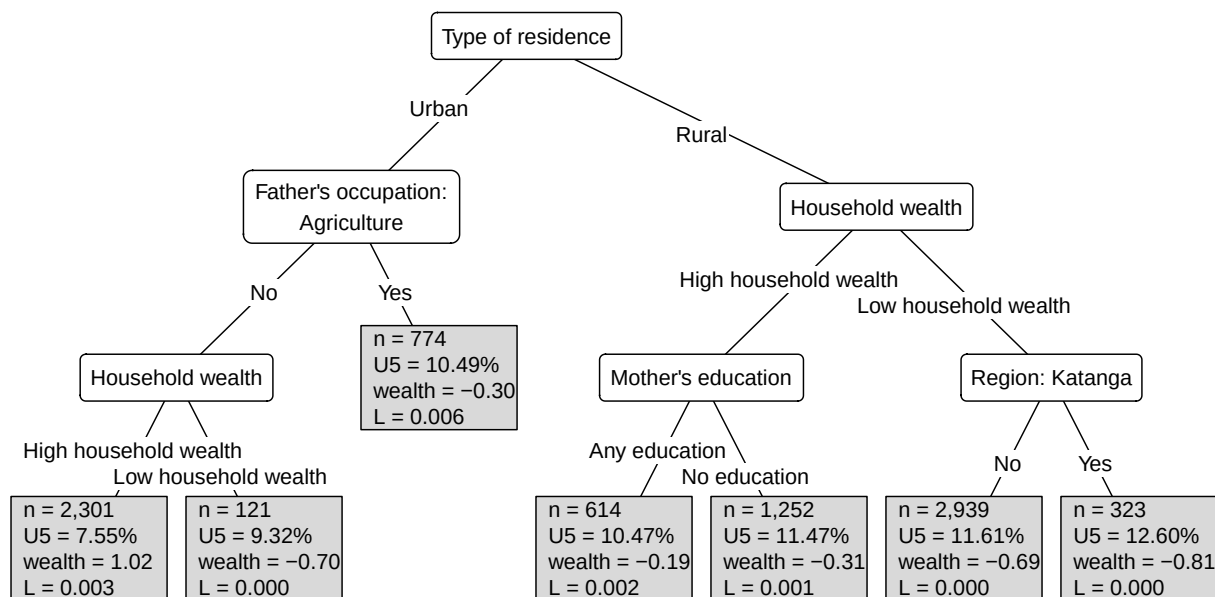


Figure 23: Surrogate tree for the L concentration-index forest.

### 5.1.7 Appendix S: Simulation Sensitivity Plots

The Clg tree is shown in the main results section. The remaining positive-control simulation trees are shown here to check whether the recovered subgroup pattern is stable across concentration-index criteria.

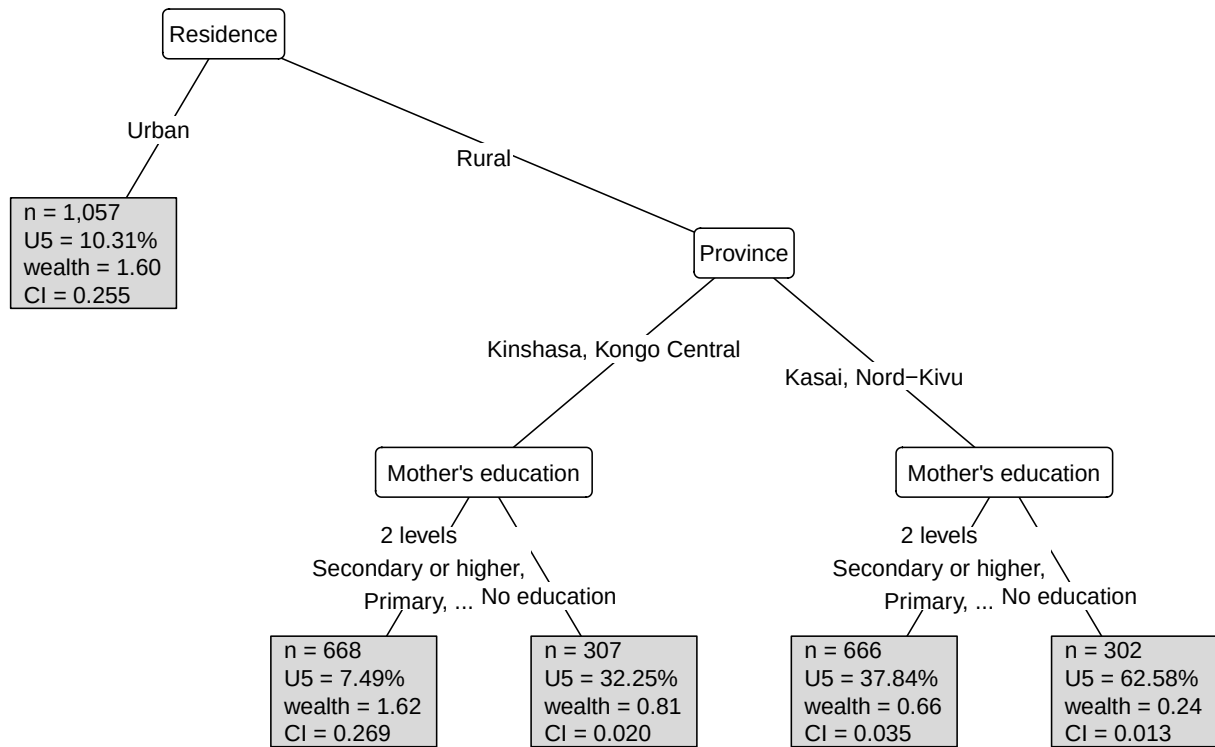


Figure 24: Positive-control simulation tree fitted with the CI criterion.

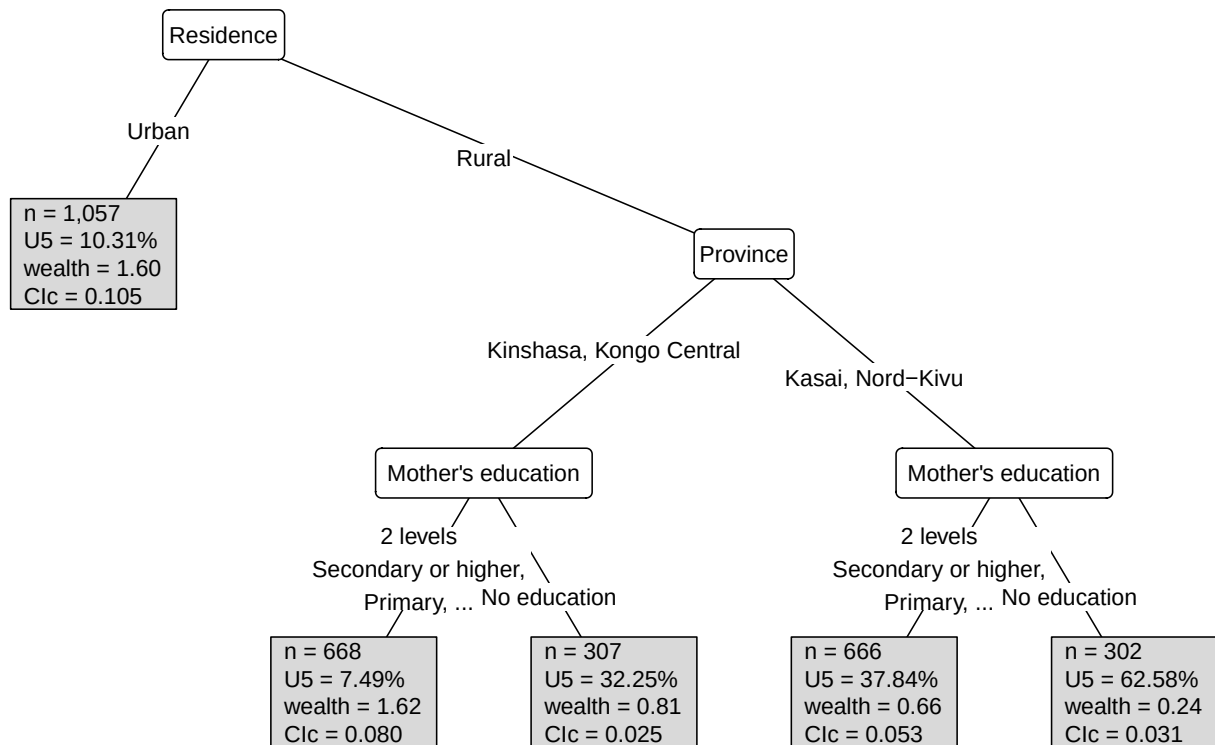


Figure 25: Positive-control simulation tree fitted with the CIc criterion.

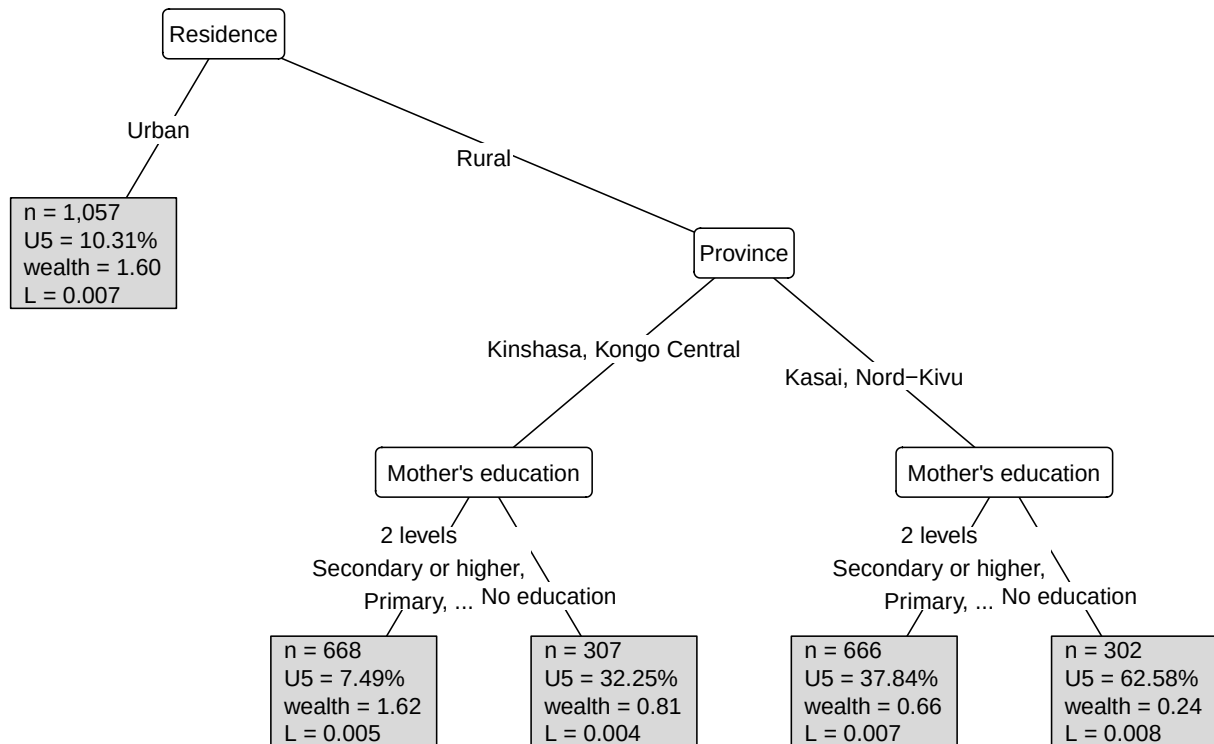


Figure 26: Positive-control simulation tree fitted with the L criterion.

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